Model Validation Fundamentals: Assumptions, Metrics, Best Practices

Table of Contents

Table of Contents 1
1. Introduction: Why Assumption Checking Matters 2
The Problem 2
Regulatory Context 3
Regulation Key Requirement 3
Why This Matters Practically 3
2. Model Types and Their Assumptions 4
A. Linear Regression (for Continuous Outcomes) 4
B. Logistic Regression (for Binary Outcomes - PD Models) 5
Key Metrics for Logistic Regression: 6
C. ARIMA / Time Series Models (Macroeconomic Scenarios) 6
Key Metrics for Time Series: 7
3. Dataset Considerations 7
Common Challenges for Credit Risk models 7

The "Merged Dataset" Problem (Independence Violation) 8
INCORRECT: Merging fiscal year rows for same borrower 8
CORRECT: Aggregate to obligor-level (loan-level) with lifetime outcome 9
Why This Matters: 9
4. Generic Implementation Guide (Python) 9
5. References & Further Reading 14
Regulatory Sources 14
Technical References 15
Accessible Free Resources 15
Disclaimer & Usage Notice 16

1. Introduction: Why Assumption Checking Matters

The Problem

In regulated industries (banking, healthcare, insurance), model validation often focuses exclusively on performance thresholds (e.g., AUC > 0.70, KS > 30). However, violated assumptions can invalidate model outputs regardless of performance metrics.

Regulatory Context

Regulation Key Requirement

SR 11-7 (Federal Reserve)	"Model validation should assess conceptual soundness, including underlying assumptions and data quality."
Basel II/III	"Internal models must be validated for statistical soundness and ongoing monitoring."
CECL (ASC 326)	"Expected credit loss estimates require reasonable and supportable forecasts with documented assumptions."
OCC 2011-12	"Banks must establish governance around model risk, including assumption testing."
Quote from SR 11-7:	"Validation should include an evaluation of the model's conceptual soundness... including the reasonableness of underlying assumptions."

Why This Matters Practically

- False Confidence: A model can have high AUC but violated independence assumptions, leading to incorrect risk estimates.
- Regulatory Scrutiny: Examiners increasingly probe assumption testing during CCAR/ICAAP reviews.

- Business Impact: Invalid assumptions can lead to capital miscalculations, reserve errors, and compliance failures.

2. Model Types and Their Assumptions

A. Linear Regression (for Continuous Outcomes)

Used for: Loss Given Default (LGD), Exposure at Default (EAD), P&L modeling

Assumption	Description	Diagnostic Test	Acceptable Threshold
Linearity	Relationship between predictors and outcome is linear	Residual plots, Ramsey RESET test	p-value > 0.05
Independence	Observations are independent	Durbin-Watson test	1.5 – 2.5
Homoscedasticity	Constant variance of residuals	Breusch-Pagan test, residual plots	p-value > 0.05

Normality	Residuals are normally distributed	Shapiro-Wilk, Q-Q plots	p-value > 0.05
No Multicollinearity	Predictors are not highly correlated	VIF (Variance Inflation Factor)	VIF < 5 (Strict)

B. Logistic Regression (for Binary Outcomes - PD Models)

Used for: Default Prediction modeling

Assumption	**Description**	**Diagnostic Test**	**Acceptable Threshold**
Linearity in Log-Odds	Predictors have linear relationship with log-odds	Box-Tidwell test, residual analysis	p-value > 0.05
Independence	Observations are independent	Study design, clustering checks	One observation per unit
No Perfect Separation	No predictor perfectly predicts outcome	Check for separation, regularization	No perfect split
Adequate Sample Size	Sufficient events per variable	EPV ≥ 10 rule of thumb	EPV ≥ 10

Correct Specification	All relevant variables included	Hosmer-Lemeshow, Brier score	p-value > 0.05 (HL), score < 0.2 (Brier)

Key Metrics for Logistic Regression:

- Hosmer-Lemeshow (HL): Calibration test (p > 0.05 = good fit)
- AUC / Gini: Discrimination (AUC > 0.70 acceptable)
- Brier Score: Overall accuracy (Lower is better)
- KS Statistic: Max separation (KS > 30 acceptable)
-

C. ARIMA / Time Series Models (Macroeconomic Scenarios)

Used for: ICAAP/CCAR Stress Testing, Interest Rate projections

Assumption	**Description**	**Diagnostic Test**	**Acceptable Threshold**
Stationarity	Mean/variance constant over time	Augmented Dickey-Fuller (ADF), KPSS	p-value < 0.05 (ADF)
No Autocorrelation	Residuals are white noise	Ljung-Box Test, ACF/PACF plots	p-value > 0.05

Normality of Residuals	Errors normally distributed	Shapiro-Wilk, Jarque-Bera	p-value > 0.05
Homoscedasticity	Constant variance of errors	ARCH-LM Test	p-value > 0.05
Model Order Selection	Optimal p, d, q parameters	AIC, BIC, AICC	Lower values preferred

Key Metrics for Time Series:

- RMSE / MAPE: Forecast accuracy
- Theil's U: Relative forecast accuracy
- Out-of-Sample Accuracy: Performance on hold-out period

3. Dataset Considerations

Common Challenges for Credit Risk models

Class Imbalance	Defaults are rare (often < 5%)	Stratified sampling, SMOTE, class weights
Time Dependency	Observations from same borrower are correlated	Aggregate to loan-level, use clustered SE

Vintage Effects	Different origination periods have different risk profiles	Vintage analysis, cohort controls
CECL Requirements	Need lifetime default estimates	Merge across fiscal years at borrower level, not row level
Data Leakage	Future information leaks into training	Strict train/test temporal split

The "Merged Dataset" Problem (Independence Violation)

When validating ICAAP/CCAR or PD models for CECL

INCORRECT: Merging fiscal year rows for same borrower

Loan_ID	**Fiscal Year (FY)**	**Predicted_PD**	**Default**
L001	20	0.02	0
L001	21	0.03	0
L001	22	0.05	1 ← Same borrower, correlated

CORRECT: Aggregate to obligor-level (loan-level) with lifetime outcome

Loan_ID	Average_ Predicted_PD	Default
L001	0.033	1
L002	0.0015	0

Why This Matters:

- Standard tests (Hosmer-Lemeshow, Chi-Square) assume independent observations.
  The same independence assumption violation also compromises the Accuracy Ratio (AR) statistics (which detects autocorrelation) and any hypothesis test applied to Brier score - although Brier score as a descriptive metric remains interpretable.
- Merged rows violate independence → inflated sample size → invalid p-values
- Solution: One row per loan with aggregated outcome flag

4. Generic Implementation Guide (Python)

Note: This section provides a conceptual framework. Full code examples are available in my GitHub repo https://github.com/kyramichel/Quant-xAI

```
"""
Model Validation Framework
    - Generic implementation - no proprietary data or methods
    - Includes checks for Linear, Logistic, and ARIMA models
"""
import pandas as pd
import numpy as np
from statsmodels.tsa.stattools import adfuller, acf, pacf
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.stats.diagnostic import acorr_ljungbox
from scipy.stats import shapiro
# ==============================================================
# 1. TIME SERIES STATIONARITY CHECK (ARIMA)
# ==============================================================
def check_stationarity(series, name="Series"):
    """
    Perform Augmented Dickey-Fuller test for stationarity.
    Returns True if stationary (p < 0.05).
    """
    result = adfuller(series.dropna(), autolag='AIC')
    print(f"\n{name} - ADF Test:")
    print(f"  ADF Statistic: {result[0]:.4f}")
    print(f"  p-value: {result[1]:.4f}")

    if result[1] < 0.05:
        print("  ✓ Stationary (Reject Null Hypothesis)")
```

```
        return True
    else:
        print("
 Non-Stationary (Fail to reject Null)")
        return False
# ============================================================
# 2. AUTOCORRELATION IN RESIDUALS (ARIMA)
# ============================================================
def check_residual_autocorrelation(residuals, lags=10):
    """
    Ljung-Box test for autocorrelation in residuals.
    Returns True if no significant autocorrelation (p > 0.05).
    """
    lb_test = acorr_ljungbox(residuals, lags=[lags], return_df=True)
    p_value = lb_test['lb_pvalue'].values[0]

    print(f"\nLjung-Box Test (Lag {lags}):")
    print(f"  p-value: {p_value:.4f}")

    if p_value > 0.05:
        print("  ✓ No significant autocorrelation in residuals")
        return True
    else:
        print("
 Significant autocorrelation detected")
        return False
# ============================================================
```

```
# 3. FULL VALIDATION WORKFLOW (Conceptual)
# ============================================================
def run_arima_validation(series, order=(1,1,1)):
    """
    End-to-end ARIMA validation workflow.
    """
    print("=" * 60)
    print("ARIMA MODEL VALIDATION WORKFLOW")
    print("=" * 60)

    # Step 1: Check Stationarity
    is_stationary = check_stationarity(series, "Original Series")
    if not is_stationary:
        print("  → Differencing required (d > 0)")

    # Step 2: Fit Model
    model = ARIMA(series, order=order)
    results = model.fit()

    # Step 3: Check Residuals
    residuals = results.resid
    check_residual_autocorrelation(residuals)
    check_normality(residuals)

    # Step 4: Forecast Accuracy
    forecast = results.forecast(steps=10)
    # Compare with actual hold-out data if available
```

```
    return results
def check_normality(residuals):
    """Shapiro-Wilk test for normality of residuals."""
    stat, p = shapiro(residuals)
    print(f"\nNormality Test (Shapiro-Wilk):")
    print(f"  p-value: {p:.4f}")
    if p > 0.05:
        print("  ✓ Residuals are normally distributed")
    else:
        print("
 Residuals deviate from normality")
# ============================================================
# MAIN EXECUTION
# ============================================================
if __name__ == "__main__":
    # Generate synthetic time series data
    np.random.seed(42)
    t = np.arange(100)
    trend = 0.1 * t
    noise = np.random.normal(0, 1, 100)
    series = trend + noise

    # Run validation
    results = run_arima_validation(series, order=(1,1,0))

    print("\n" + "=" * 60)
```

```
print("VALIDATION COMPLETE")
print("=" * 60)
```

5. References & Further Reading

Regulatory Sources

Source	Link
Federal Reserve SR 11-7	https://www.federalreserve.gov/supervisionreg/srletters/sr1107.htm
OCC Model Risk Management	https://www.occ.treas.gov/publications-and-resources/publications/bank-supervision-policies/pub-bank-supervision-policies-2011-12.html
Basel Committee on Validation	https://www.bis.org/publ/bcbs195.htm
CECL (ASC 326)	https://www.fasb.org/page/PageContent?pageId=/reference-library/accounting-standards-codification/asc-326.html

Technical References

Source	Topic
Wooldridge (2010)	Econometric Analysis of Cross Section and Panel Data
Hosmer & Lemeshow (2000)	Applied Logistic Regression
Box, Jenkins, Reinsel	Time Series Analysis: Forecasting and Control
NIST/SEMATECH e-Handbook	Statistical Methods for Validation

Accessible Free Resources

Resource	Best For	Link
Statistics How To	Hosmer-Lemeshow, Chi-Square basics	statisticshowto.com
Wikipedia	Chi-Squared Test Validity Section	wikipedia.org
LibreTexts	Linear & Logistic Regression	stats.libretexts.org

	Assumptions	
NIST Handbook	Time Series & ARIMA	itl.nist.gov

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